

A Randomized Clinical Trial Comparing Patellar Tendon, Hamstring Tendon, and Double-Bundle ACL Reconstructions

Patient-Reported and Clinical Outcomes at 5-Year Follow-up

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Background: This randomized clinical trial evaluated the 5-year, disease-specific quality-of-life outcomes of patients treated for anterior cruciate ligament (ACL) deficiency with use of a patellar tendon, single-bundle quadruple-stranded hamstring, or double-bundle hamstring reconstruction.

Methods: Patients (n = 110 per group; 183 male; 14 to 50 years of age) were randomly allocated to anatomically positioned ACL reconstruction with use of a patellar tendon graft (patellar tendon group; mean age, 28.7 years), quadruple-stranded hamstring tendon construct (hamstring tendon group; mean age, 28.5 years), or double-bundle hamstring tendon construct (double-bundle group; mean age, 28.3 years). Computer-generated allocation with varied block randomization was performed intraoperatively. The 5-year primary outcome measure was the Anterior Cruciate Ligament-Quality of Life (ACL-QOL) score, and secondary outcome measures included the International Knee Documentation Committee (IKDC) subjective score and objective grades, pivot shift assessment, range of motion, kneeling pain, Tegner activity scale, Cincinnati Occupational Rating Scale, complete traumatic ruptures, partial traumatic tears, total traumatic reinjuries, and atraumatic graft failures.

Results: Three hundred and fifteen patients (95%) completed the 5-year follow-up (4 withdrawals, and 11 lost to follow-up). ACL-QOL scores increased significantly from baseline for all groups ($p < 0.0001$), but mean scores at 5 years did not differ among the groups ($p = 0.548$): mean (and standard deviation) of 82.5 ± 17.9 (95% confidence interval [CI], 79.0 to 86.0) for patellar tendon, 83.9 ± 18.2 (95% CI, 80.3 to 87.4) for hamstring tendon, and 81.1 ± 19.3 (95% CI, 77.4 to 84.8) for double-bundle. The proportions of patients at 5 years with a pivot shift grade of ≥ 2 (12% for patellar tendon, 16% for hamstring tendon, and 22% for double-bundle; $p = 0.149$) and with normal and nearly normal knees per IKDC grade (87%, 82%, and 75% for the 3 groups, respectively; $p = 0.093$) trended in favor of patellar tendon reconstruction. No differences were seen for the IKDC subjective scores, knee range of motion including passive extension deficits, the single leg hop test, Cincinnati Occupational Rating scores, and Tegner activity levels. Kneeling pain was more common with patellar tendon reconstruction (10% of patients compared with 4% for hamstring tendon and 2% for double-bundle; $p = 0.029$). For hamstring tendon and double-bundle reconstructions, there were trends toward more complete traumatic ruptures (4% for patellar tendon, 10% for hamstring tendon, and 10% for double-bundle; $p = 0.145$) and partial traumatic tears (0%, 5%, and 6%, respectively; $p = 0.059$), with significantly fewer combined traumatic reinjuries in the patellar tendon group (4 patients compared with 16 for hamstring tendon and 17 for double-bundle; $p = 0.010$). Atraumatic graft failure (pivot shift grade of ≥ 2) was found for 10% of the patients.

Conclusions: At 5 years, we found no significant difference in the quality-of-life outcome among the patellar tendon, hamstring tendon, and double-bundle techniques for ACL reconstruction. Significantly more patients in the hamstring tendon and double-bundle groups experienced traumatic graft reinjury compared with the patellar tendon group. There was

continued

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a trend toward a higher percentage with normal and nearly normal IKDC grades in the patellar tendon group compared with hamstring tendon and double-bundle groups.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

Anterior cruciate ligament (ACL) injuries affect quality of life¹ and knee function of thousands of young people annually^{2,3}, typically those in their prime⁴⁻⁶. Since the popularization of the double-bundle ACL reconstruction technique, several randomized clinical trials evaluating associated outcomes have been published. The double-bundle technique is intended to reconstruct the 2-bundle anatomy of the ACL, thereby restoring biomechanical function, including knee rotational stability^{7,8}. The short-term clinical outcomes of the published randomized trials comparing single and double-bundle ACL reconstruction have been analyzed in at least 3 separate meta-analyses⁹⁻¹¹ and 1 Cochrane systematic review¹². The Cochrane review concluded that there was insufficient evidence to determine the superiority of single versus double-bundle reconstruction and acknowledged the need for high-quality, large randomized clinical trials with long-term follow-up to assess reinjuries, graft failure, revision surgery, and osteoarthritis¹².

Intermediate clinical outcomes of the use of autograft tendons (5 to 8 years postoperatively) have been reported in only a small number of randomized clinical trials¹³⁻²¹. Three studies compared the use of patellar-tendon and single-bundle hamstring tendon ACL reconstructions¹³⁻¹⁵. Six other studies compared single and double-bundle hamstring reconstructions¹⁶⁻²¹ using small sample sizes that ranged from 31²⁰ to 87²¹ subjects.

We previously reported the 2-year outcomes of the largest, to our knowledge, double-blinded randomized clinical trial comparing the use of patellar tendon, semitendinosus + gracilis tendon, and double-bundle semitendinosus + gracilis tendon reconstruction techniques for primary ACL deficiency;

330 patients were included^{22,23}. This article presents the 5-year, disease-specific Anterior Cruciate Ligament Quality of Life (ACL-QOL) outcomes and clinical outcomes from the same randomized clinical trial.

Materials and Methods

The University of Calgary Conjoint Health Research Ethics Board approved this randomized clinical trial (ClinicalTrials.gov number NCT00529958). The trial methodology, sample-size calculations, and surgical techniques were previously described²², as were complications and adverse events²⁴, 2-year clinical and functional outcomes²², and reruptures and reinjuries²³.

In summary, of 745 patients screened at the University of Calgary Sport Medicine Centre between November 2007 and December 2010, a total of 439 patients with isolated ACL deficiency were eligible for inclusion (Table I). All patients provided written informed consent to participate. Intraoperative examination under anesthesia and diagnostic arthroscopy further confirmed study eligibility (Fig. 1).

Using computer-based, varied block randomization with allocation concealment, 330 eligible patients were randomly assigned intraoperatively to ACL reconstruction with use of 1 of 3 techniques: a patellar tendon graft (the patellar tendon group), a hamstring tendon graft using semitendinosus and gracilis tendons in a single-bundle, quadruple-stranded construct (the hamstring tendon group), or a 2-stranded semitendinosus and 2-stranded gracilis tendon double-bundle construct (the double-bundle group). As previously described²², the operative procedures for each ACL reconstruction technique were the same, with the exception of

TABLE I Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
(1) Confirmed diagnosis of ACL deficiency based on all of the following: (a) History of traumatic injury episode; (b) Physical examination findings of increased anterior translation of the tibia on the femur (Lachman test and/or anterior drawer test); (c) Positive pivot-shift test; (d) Radiographs showing skeletal maturity (i.e., tibial tubercle fused) and no fractures (n.b., magnetic resonance imaging [MRI] not required) (2) Age of 14-50 yr	(1) Combined ligament deficiencies (posterior cruciate, medial and/or lateral collateral deficiency; n.b., grade-1 side-to-side differences [i.e., <5 mm] on valgus, varus, or posterior stress testing were not considered exclusions) (2) Intraoperative identification of International Cartilage Repair Society (ICRS) grade-4 chondral lesion >1 cm ² in size ²⁵ (3) Previous ligament surgery on the affected or contralateral knee (4) Cases involving litigation or Workers' Compensation (5) Confirmed connective-tissue disorder (e.g., Ehlers-Danlos syndrome, Marfan syndrome) (6) Inability to speak, read, or understand the English language

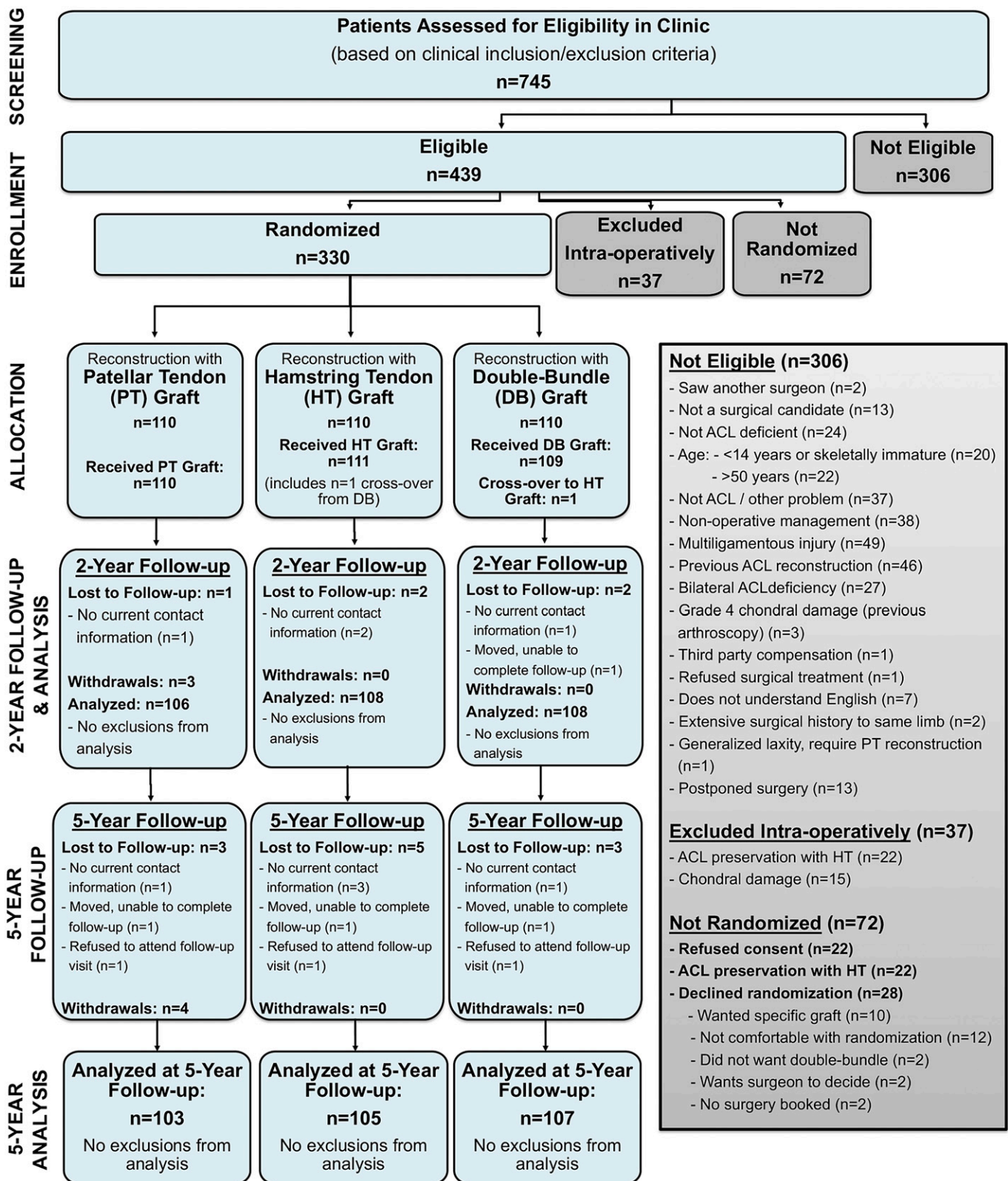


Fig. 1.

CONSORT (Consolidated Standards of Reporting Trials) diagram showing the flow of patients screened, excluded, randomized, followed, and analyzed in the trial.

TABLE II Demographic Characteristics of Patients with 5-Year Follow-up

Characteristic	Patellar Tendon (N = 103)	Hamstring Tendon (N = 105)	Double-Bundle (N = 107)	P Value
Age at 5-yr follow-up* (yr)	33.8 ± 9.8 (31.9-35.7)	33.7 ± 10.0 (31.7-35.6)	33.2 ± 9.8 (31.3-35.09)	0.907
Sex				0.903
Male	60	58	60	
Female	43	47	47	
Tear				0.993
Acute	16	16	16	
Chronic	87	89	91	
Side				0.184
Left	48	61	51	
Right	55	44	56	
Follow-up duration* (mo)	59.8 ± 2.0 (59.4-60.2)	60.0 ± 3.1 (59.3-60.6)	59.8 ± 1.8 (59.4-60.1)	0.809

*The values are given as the mean and standard deviation, with the 95% confidence interval in parentheses.

the autograft tissue type, and the utilization of 2 tibial and femoral tunnels for the double-bundle technique. All grafts were placed in the center of the true anatomic footprints on the tibial and femoral ACL attachment sites. All procedures used an endoscopic 1-incision technique, with a universal 5 to 7-cm anteromedial vertical incision for graft harvesting, EndoButton (Smith & Nephew) suspensory fixation on the femoral side, and bioabsorbable interference screw (BioScrew; Linvatec) fixation on the tibial side. Follow-up and rehabilitation protocols²⁵ were identical for all groups. The patient and inde-

pendent trained examiner remained blinded to treatment group allocation until the 2-year follow-up period.

Study visits at the University of Calgary Sport Medicine Centre occurred preoperatively (baseline) and postoperatively at 1, 2, and 5 years. At each visit, the examiner conducted a standardized bilateral physical knee examination and tests, including anterior and posterior drawer test, Lachman test, and pivot shift test (equal = 0, glide = 1, clunk = 2, and gross = 3); passive range-of-motion assessment (flexion and extension, in degrees) using a long-arm goniometer; and assessment of

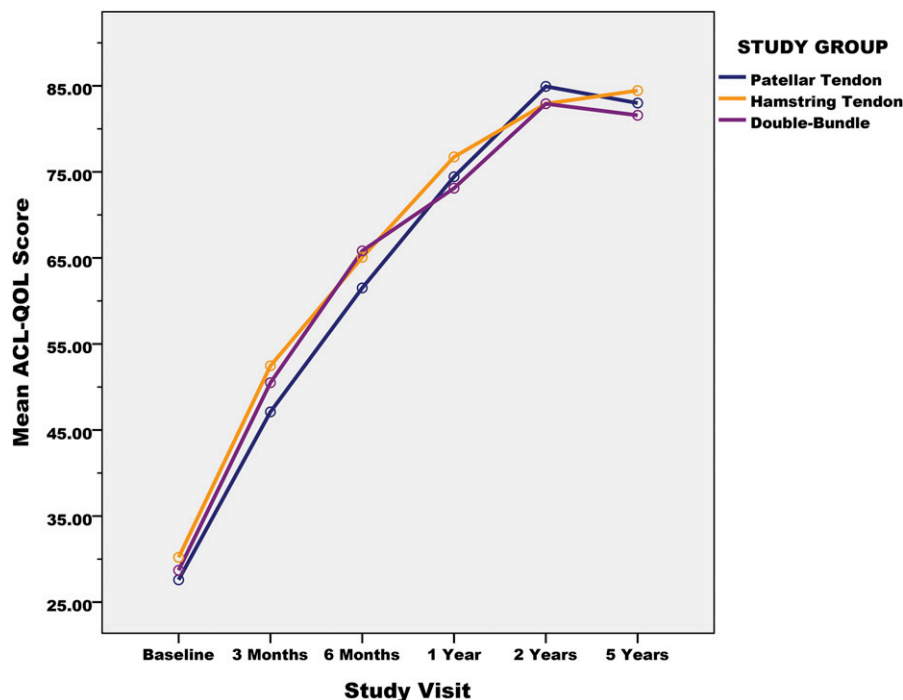


Fig. 2.

Change in ACL-QOL scores from baseline to 5 years postoperatively. ACL-QOL = Anterior Cruciate Ligament-Quality of Life.

TABLE III ACL-QOL Scores Over Time*

Group	Baseline	Postop.		
		1 Yr	2 Yr	5 Yr
Patellar tendon	N = 110	N = 103	N = 106	N = 103
Mean and SD	27.0 ± 13.1	74.5 ± 18.7	84.6 ± 16.6	82.5 ± 17.9
95% CI	24.5-29.5	70.9-78.2	81.4-87.8	79.0-86.0
Hamstring tendon	N = 110	N = 107	N = 108	N = 105
Mean and SD	29.8 ± 14.7	75.4 ± 18.5	82.5 ± 17.7	83.9 ± 18.2
95% CI	27.0-32.6	71.9-79.0	79.2-85.9	80.3-87.4
Double-bundle	N = 110	N = 107	N = 108	N = 107
Mean and SD	28.7 ± 13.7	73.6 ± 19.2	82.4 ± 17.5	81.1 ± 19.3
95% CI	26.1-31.3	69.9-77.3	79.1-85.7	77.4-84.8
P value	0.324	0.779	0.591	0.548

*SD = standard deviation, and CI = confidence interval.

kneeling pain (mild, moderate, or severe). The primary outcome measure was the patient-generated and reported ACL-QOL¹, a 32-item, disease-specific questionnaire that assesses symptoms or physical complaints and work-related, sports/recreational, lifestyle, and social/emotional concerns. The instrument uses a response format of a 0 to 100-mm visual analog scale, with a higher score representing better quality of life¹. Secondary outcome measures included the Tegner activity scale²⁶, the International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form

(maximum score of 100)^{27,28} and Objective Group grade (normal, nearly normal, abnormal, or severely abnormal)^{29,30}, the Cincinnati Occupational Rating Scale (maximum score of 100)³¹, and the single leg hop test (percent of opposite side)²⁹. The proportions of traumatic reinjuries (complete traumatic reruptures and partial traumatic tears combined), atraumatic graft failures, and contralateral ACL ruptures were determined. *Complete traumatic reruptures* were defined as a consequence of an acute traumatic event resulting in an obvious change in static stability (definite loss of end point on

TABLE IV Pivot Shift Measurements Over Time*

Group	Baseline	Postop.		
		1 Yr	2 Yr	5 Yr
Patellar tendon	N = 110	N = 101	N = 102	N = 99
Equal: 0	0 (0)	47 (47)	36 (35)	37 (37)
Glide: 1	6 (5)	51 (50)	52 (51)	50 (51)
Clunk: 2	103 (94)	3 (3)	14 (14)	12 (12)
Gross: 3	1 (1)	0 (0)	0 (0)	0 (0)
Hamstring tendon	N = 110	N = 106	N = 104	N = 99
Equal: 0	1 (1)	46 (43)	28 (27)	29 (29)
Glide: 1	6 (5)	48 (45)	57 (55)	54 (55)
Clunk: 2	102 (93)	12 (11)	18 (17)	16 (16)
Gross: 3	1 (1)	0 (0)	1 (1)	0 (0)
Double-bundle	N = 110	N = 102	N = 107	N = 103
Equal: 0	0 (0)	53 (52)	38 (35)	30 (29)
Glide: 1	9 (8)	41 (40)	49 (46)	50 (49)
Clunk: 2	99 (90)	7 (7)	19 (18)	22 (21)
Gross: 3	2 (2)	1 (1)	1 (1)	1 (1)

*The values are given as the number of patients, with the percentage in parentheses.

TABLE V IKDC Objective Grade Over Time

Group	No. of Patients			
	Normal (A)	Nearly Normal (B)	Abnormal (C)	Severely Abnormal (D)
Baseline (p = 0.595)				
Patellar tendon (n = 109)	0	1	103	5
Hamstring tendon (n = 109)	0	4	101	4
Double-bundle (n = 110)	0	5	100	5
1 yr postop. (p = 0.461)				
Patellar tendon (n = 101)	9	79	13	0
Hamstring tendon (n = 104)	9	77	18	0
Double-bundle (n = 101)	9	71	19	2
2 yr postop. (p = 0.658)				
Patellar tendon (n = 101)	12	65	23	1
Hamstring tendon (n = 104)	12	62	28	2
Double-bundle (n = 107)	19	56	30	2
5 yr postop. (p = 0.215)				
Patellar tendon (n = 98)	24	61	13	0
Hamstring tendon (n = 99)	18	63	18	0
Double-bundle (n = 103)	19	58	24	2

Lachman testing, increased anterior translation [>3 mm], and pivot shift grade of ≥ 2) and confirmed by magnetic resonance imaging (MRI) or diagnostic arthroscopy. Revision ACL reconstruction was recommended for all patients with a complete traumatic rerupture. *Partial traumatic tears* were defined as a consequence of an acute traumatic event resulting in a suspected meniscal injury or graft tear, without the clinical characteristics of a complete rerupture, as defined above. A partial traumatic tear was confirmed by MRI or diagnostic arthroscopy. Revision ACL reconstruction was not recommended or required. *Atraumatic*

graft failures were defined in the absence of an acute traumatic event, but with a pivot shift grade of ≥ 2 .

Statistical Analyses

The data were analyzed on an intention-to-treat basis, using a 5% significance level. For the continuous variables, adjusted Bonferroni comparisons and repeated-measures analyses, using a mixed-model analysis of variance (ANOVA) for treatment group over time of assessment, were performed. For categorical data, chi-square analyses were used.

TABLE VI IKDC Subjective Scores Over Time*

Group	Baseline	Postop.		
		1 Yr	2 Yr	5 Yr
Patellar tendon	N = 110	N = 101	N = 105	N = 103
Mean and SD	50.1 $\pm$ 13.9	79.7 $\pm$ 11.7	84.6 $\pm$ 13.8	83.9 $\pm$ 12.9
95% CI	47.5-52.8	77.4-82.0	81.9-87.3	81.4-86.5
Hamstring tendon	N = 110	N = 105	N = 107	N = 105
Mean and SD	51.0 $\pm$ 16.2	81.7 $\pm$ 12.6	85.3 $\pm$ 11.6	85.2 $\pm$ 13.0
95% CI	48.0-54.1	79.2-84.1	83.1-87.5	82.7-87.7
Double-bundle	N = 110	N = 107	N = 107	N = 107
Mean and SD	51.5 $\pm$ 14.6	80.0 $\pm$ 11.9	84.2 $\pm$ 11.8	84.3 $\pm$ 13.4
95% CI	48.7-54.3	77.7-82.3	82.0-86.5	81.7-86.9
P value	0.789	0.444	0.821	0.770

*SD = standard deviation, and CI = confidence interval.

TABLE VII Range-of-Motion, Single Leg Hop Test, and Cincinnati Occupational Rating Scale Outcomes at Baseline and 5 Years*

Outcome	Patellar Tendon	Hamstring Tendon	Double-Bundle	P Value
Passive extension deficit (no. with deficit of >3°)				
Baseline	17 (n = 110)	19 (n = 110)	23 (n = 110)	0.561
5 yr	0 (n = 98)	0 (n = 98)	0 (n = 103)	Not calculable
Passive flexion of operative knee (°)				
Baseline	N = 110	N = 110	N = 110	0.748
Mean and SD	144.4 ± 7.5	144.1 ± 7.2	143.7 ± 7.4	
95% CI	143.0-145.8	142.8-145.5	142.3-145.1	
5 yr	N = 98	N = 99	N = 103	0.275
Mean and SD	142.5 ± 7.3	142.2 ± 7.8	140.9 ± 7.8	
95% CI	141.1-144.0	140.6-143.7	139.4-142.4	
Single leg hop (% of nonoperative side)				
Baseline	N = 105	N = 107	N = 109	0.558
Mean and SD	70.7 ± 33.4	67.1 ± 33.6	71.6 ± 31.3	
95% CI	64.2-77.2	60.6-73.5	65.7-77.6	
5 yr	N = 93	N = 81	N = 84	0.194
Mean and SD	100.6 ± 15.2	94.2 ± 32.1	94.8 ± 28.2	
95% CI	97.4-103.7	87.1-101.3	88.7-100.9	
Cincinnati Occupational Rating (of 100)				
Baseline	N = 110	N = 110	N = 110	0.888
Mean and SD	36.6 ± 19.9	35.4 ± 21.3	36.3 ± 17.7	
95% CI	32.8-40.4	31.4-39.4	32.9-39.6	
5 yr	N = 103	N = 105	N = 107	0.813
Mean and SD	35.9 ± 18.4	36.6 ± 23.8	37.8 ± 24.2	
95% CI	32.3-39.5	32.0-41.2	33.2-42.5	

*SD = standard deviation, and CI = confidence interval.

In order to evaluate whether patients with reinjury and/or a contralateral tear would negatively bias the outcomes, subgroup analyses of ACL-QOL scores, pivot shift outcomes, IKDC objective grades, and Tegner activity levels were performed. The subgroup analyses for the ACL-QOL, IKDC objective grade, and Tegner activity level outcomes excluded patients with traumatic reinjury and/or contralateral ACL tear. A subgroup analysis comparing graft size for patients with and without reinjury was also performed. IBM SPSS Statistics software (version 25; IBM) was used.

Results

Patients (n = 110 per group; 183 male; 14 to 50 years of age) were randomly allocated to anatomically positioned ACL reconstruction with use of a patellar tendon graft (patellar tendon group; mean age, 28.7 years), quadruple-stranded hamstring tendon (hamstring tendon group; mean age, 28.5 years), or double-bundle hamstring tendon (double-bundle group; mean age, 28.3 years).

A total of 315 (95%) of the original 330 randomized patients completed 5-year follow-up between 2012 and 2016; there were no differences among the groups with respect to demographic data (Table II). Cumulatively, 4 patients in the patellar tendon group withdrew, and 11 patients were lost to 5-year follow-up (3 in the patellar tendon group, 5 in the hamstring tendon group, and 3 in the double-bundle group) (Fig. 1).

Mean ACL-QOL scores increased significantly from baseline to 5 years for all groups ($p < 0.0001$). The changes from baseline to 5 years and from 2 to 5 years (Fig. 2) did not differ significantly among the groups ($p = 0.619$ and $p = 0.355$, respectively). Mean 5-year ACL-QOL scores also did not differ among the groups ($p = 0.548$) (Table III).

Pivot shift grades at all time periods were similar among the groups (Table IV). However, the proportions of patients with a pivot shift grade of ≥ 2 at 5 years demonstrated a trend in favor of patellar tendon reconstruction (12% compared with 16% for hamstring tendon and 22% for double-bundle; $p = 0.149$).

TABLE VIII Proportion of Patients Returning to Preinjury Tegner Activity Level and with Tegner Activity Level of ≥ 5

	No. of Patients/Total No. (%)			P Value
	Patellar Tendon	Hamstring Tendon	Double-Bundle	
Return to preinjury level*				
1 yr	27/101 (27)	33/105 (31)	28/107 (26)	0.648
2 yr	44/105 (42)	43/107 (40)	36/107 (34)	0.426
5 yr	35/103 (34)	41/105 (39)	41/107 (38)	0.716
Return to preinjury level†				
1 yr	27/100 (27)	33/101 (33)	27/101 (27)	0.575
2 yr	42/99 (42)	42/98 (43)	35/96 (37)	0.599
5 yr	33/89 (37)	33/80 (41)	35/81 (43)	0.552
Tegner activity level ≥ 5 *				
Preinjury	106/110 (96)	106/110 (96)	104/110 (95)	0.742
Baseline	12/110 (11)	19/110 (17)	20/110 (18)	0.267
1 yr	77/101 (76)	79/105 (75)	79/107 (74)	0.922
2 yr	86/103 (84)	86/107 (80)	88/106 (83)	0.814
5 yr	80/103 (78)	84/105 (80)	86/107 (80)	0.872
Tegner activity level ≥ 5 †				
1 yr	76/100 (76)	77/101 (76)	74/101 (73)	0.863
2 yr	80/97 (83)	80/98 (82)	81/95 (85)	0.781
5 yr	70/89 (79)	66/80 (83)	67/81 (83)	0.699

*No exclusion of patients with traumatic reinjuries and contralateral ACL tears. †Excluding patients with traumatic reinjuries and contralateral ACL tears.

IKDC Objective Group grades were similar among the groups at all time periods (Table V). At 5 years, there was a trend in favor of the patellar tendon group, with a higher percentage of knees graded normal and nearly normal: 87% compared with 82% for hamstring tendon and 75% for double-bundle ($p = 0.093$).

IKDC subjective scores (Table VI) increased significantly from baseline to 5 years for all groups ($p < 0.0001$). The mean change in IKDC subjective scores from 2 to 5 years did not differ significantly among the groups ($p = 0.976$).

No patient had a passive extension deficit of $>3^\circ$ at 5-year follow-up (Table VII). There was a minor change from baseline

in passive flexion range of motion, but there were no statistically significant between-group differences (Table VII). In regard to the single leg hop test, the patellar tendon group demonstrated the highest mean percentage relative to the contralateral knee at 5 years, but the difference between groups was not significant ($p = 0.194$) and likely not clinically important. Cincinnati Occupational Rating scores remained consistent relative to baseline at the 5-year follow-up, with no significant difference among the groups ($p = 0.813$) (Table VII).

There was no difference among the groups at 5 years in the proportion of patients with a Tegner activity level of ≥ 5 ($p = 0.872$) or who had returned to the preinjury activity level ($p = 0.716$) (Table VIII). The proportion of patients in each group returning to the preinjury activity level increased from 1 to 2 years (Table VIII), but from 2 to 5 years, the proportion decreased for the patellar tendon group (to 34% at 5 years), remained consistent for the hamstring tendon group (39%), and increased for the double-bundle group (38%). When excluding patients with traumatic reinjuries and contralateral ACL tears, a similar trend was shown (37% in the patellar tendon group, 41% in the hamstring tendon group, and 43% in the double-bundle group), which did not reach significance (Table VIII).

A significantly greater proportion of patients with patellar tendon reconstructions reported moderate to severe kneeling pain at 5 years ($p = 0.029$) (Table IX).

Twenty-six (8.3%) of the patients overall sustained a complete traumatic rerupture, 11 (3.5%) experienced a partial

TABLE IX Proportion of Patients Reporting Moderate or Severe Kneeling Pain

	Baseline	2 Yr	5 Yr
Patellar tendon	27% (29/109)	17% (17/98)	10% (10/98)
Hamstring tendon	26% (29/110)	9% (9/103)	4% (4/98)
Double-bundle	16% (18/110)	4% (4/104)	2% (2/101)
P value	0.121	0.005*	0.029*

*Significant ($p \leq 0.05$).

TABLE X Proportion of Patients with and without Traumatic Reinjuries, and with Contralateral ACL Tears at 5 Years Post-ACL Reconstruction

	Patellar Tendon (N = 103)	Hamstring Tendon (N = 105)	Double-Bundle (N = 107)	P Value
No traumatic reinjury to index knee (<i>no. [%]</i>)	99 (96%)	89 (85%)	90 (84%)	
Total traumatic reinjuries (<i>no. [%]</i>)	4 (4%)	16 (15%)	17 (16%)	0.010*
Complete traumatic rerupture	4 (4%)	11 (10%)	11 (10%)	0.145
Partial traumatic tear	0 (0%)	5 (5%)	6 (6%)	0.059
Contralateral ACL tear (<i>no. [%]</i>)	10 (10%)	9 (9%)	9 (8%)	0.938

*Significant ($p \leq 0.05$).

traumatic tear, and 28 (8.9%) had a contralateral ACL tear. There was no difference among the groups in the proportion of patients with contralateral ACL tears (Table X). Thirty-seven (11.7%) of the patients overall had a traumatic reinjury; significantly fewer patients in the patellar tendon group experienced reinjury compared with the hamstring tendon and double-bundle groups ($p = 0.010$) (Table X). When the traumatic reinjuries were subclassified as complete traumatic reruptures and partial traumatic tears, the differences were no longer statistically significant, but trended in favor of patellar tendon reconstruction (Table X). Revision ACL surgery was performed in 22 (5 one-stage and 17 two-stage) of the 26 patients with complete traumatic rerupture (3 in the patellar tendon group, 10 in the hamstring tendon group, and 9 in the double-bundle group). Four patients (1 patellar tendon, 1 hamstring tendon, and 2 double-bundle) chose not to undergo revision surgery. Between 2 and 5-year follow-up, 7 additional patients required arthroscopic surgery on the index knee: 1 patient in the double-bundle group had a hypertrophied graft with notch impingement; 4 patients re-tore a previously repaired medial meniscus, requiring partial meniscectomy (2 in the hamstring tendon group and 2 in the double-bundle group); 1 patient in the double-bundle group re-tore a previously re-

paired medial meniscus in addition to sustaining a tear in the posterolateral bundle of the graft with an associated cyclops lesion; and 1 patient in the double-bundle group had a degenerative lateral meniscal tear requiring partial meniscectomy.

Thirty (10.0%) of the patients experienced an atraumatic graft failure (Table XI), based on a pivot shift grade of ≥ 2 , exclusive of those with traumatic reinjuries.

Subgroup Analyses

The subgroup analysis showed that, relative to the entire study sample (Table III), the 5-year ACL-QOL scores for the patients without injury to either knee were slightly higher (mean [and standard deviation], 84.5 ± 16.9 in the patellar tendon group, 86.9 ± 14.5 in the hamstring tendon group, 84.8 ± 14.6 in the double-bundle group), but these scores did not differ significantly between the 3 groups ($p = 0.543$).

Among the patients without reinjury, the proportion with a pivot shift grade of ≥ 2 at 5 years was lower in all 3 groups (9% for patellar tendon, 11% for hamstring tendon, and 17% for double-bundle; $p = 0.232$) compared with the group proportions for the entire study sample (12% for patellar tendon, 16% for hamstring tendon, and 22% for double-bundle) (Table IV).

TABLE XI Atraumatic Graft Failures (AGFs) and ACL-QOL Scores at 5-Year Follow-up

	Patellar Tendon (N = 99)	Hamstring Tendon (N = 98)	Double-Bundle (N = 103)
No. (%) of patients with AGF	7 (7%)	9 (9%)	14 (14%)
No. of patients without AGF	88	73	72
No. of patients excluded due to traumatic reinjury	4	16	17
ACL-QOL score of AGFs*			
Mean	68.7	89.9	85.2
SD	18.5	7.9	16.7
95% CI	51.6-85.8	83.8-96.0	75.5-94.9
Median	70.3	88.8	92.6
Range	45.6-86.5	77.4-99.5	42.3-99.9

*SD = standard deviation, and CI = confidence interval.

TABLE XII Graft Size in Patients with and without Traumatic Reinjury

	Patellar Tendon	Hamstring Tendon	Double-Bundle
Without traumatic reinjury	N = 99	N = 90*	N = 89
Average graft size (mm)	10.0	7.85	
Anteromedial bundle			6.5
Posterolateral bundle			5.3
With traumatic reinjury			
Partial graft tear	N = 0	N = 5	N = 6
Complete graft rerupture	N = 4	N = 11	N = 11
Average graft size (mm)	9.75	8.0	
Anteromedial bundle			6.5
Posterolateral bundle			5.2

*Includes 1 cross-over from the double-bundle group.

The IKDC Objective Group grades for patients with successful ACL reconstruction demonstrated proportions of abnormal and severely abnormal grades at baseline (99% in the patellar tendon group, 95% in the hamstring tendon group, and 95% in the double-bundle group; $p = 0.307$) that were comparable with those for the entire study sample (99% for patellar tendon, 96% for hamstring tendon, and 95% for double-bundle) (Table V). At the 5-year follow-up, higher proportions of uninjured patients (92% for patellar tendon, 87% for hamstring tendon, and 80% for double-bundle; $p = 0.095$) had grades of normal and nearly normal compared with proportions for the entire study sample (87% in the patellar tendon group, 82% in the hamstring tendon group, and 75% in the double-bundle group) (Table V).

Table XII presents the 5-year information on the representative graft sizes of the patients with traumatic reinjury compared with those without reinjury. The within-group comparisons demonstrated no clinically important or statistically significant differences with respect to the size of the grafts.

Discussion

This large, randomized clinical trial demonstrated that the results were durable over time (5 years). Essentially, all outcomes were similar to those of 2-year follow-up that were previously reported^{12,24}. There were consistently high patient-reported quality-of-life scores at 2 and 5 years, in all study groups (Fig. 2).

Traumatic reinjury to the graft (11.7%) and/or contralateral ACL tears (8.9%) remain the most notable problems following ACL reconstructive surgery. The number of complete and partial traumatic reinjuries combined was significantly higher in the hamstring tendon and double-bundle groups compared with the patellar tendon group. Contralateral ACL tears almost doubled from 5.2% at 2 years to 8.9% at 5 years. Most patients continued to be highly active, with roughly 80% functioning at a Tegner activity level of ≥ 5 . However, at 2 years, only 39% of the patients overall had returned to preinjury activity levels. This overall percentage was similar at 5 years

(37%), suggesting that activity levels remained stable. Patients may be protecting their knees from reinjury by reducing their activity level, or there may be a natural reduction in physical activity over time. Patients may also be participating in alternative activities to those that caused the initial injury.

The identification of what was defined as an atraumatic graft failure (i.e., a pivot shift grade of ≥ 2) remains a surgical concern. However, patients seem to adapt to this static rotational abnormality, as reflected by their relatively high subjective ACL-QOL scores. The explanation for the lack of association between “objective” and subjective information continues to be an enigma.

The higher reinjury rates in the hamstring and double-bundle groups could be explained in several ways. It is possible that using hamstring autograft tendons compromises the dynamic control of the knee, which increases the risk of reinjury. In a similar vein, it is possible that utilizing the patellar tendon provides a protective effect by weakening the quadriceps, thus creating a more favorable quadriceps-to-hamstring ratio³². However, the single leg hop test results, a proxy for muscular function, would suggest otherwise. It is also possible that the subtle variances in technique or graft size increase the risk of reinjury. The 5-year information on the representative graft sizes of patients with reinjury, compared with those without, was virtually identical (Table XII). This suggests that graft size was not predictive of reinjury in this study.

It is intuitive to suggest that an individual who has experienced an ACL tear would be inherently at higher risk for further knee injury. Whether this is because of a genetic predisposition, the patient's environment (i.e., sports participation), or, most likely, multifactorial reasons, is difficult to determine. In our study, there was no equivalent comparative group who did not have surgery or was not injured. Younger age was predictive of reinjury at the 2-year period, as was previously reported²³. Since the patients at 5-year follow-up were older and less active, and most of the reinjuries occurred within the first 2 years postoperatively, a regression analysis was not repeated at 5-year follow-up.

The proportion of patients with moderate or severe kneeling pain remained higher for the patellar tendon group, but the percentage decreased to 10% compared with 17% at 2 years²⁴. No patient had a passive extension deficit $>3^{\circ}$ at 5 years. The difference in the proportion of patients with IKDC objective knee grades of normal and nearly normal approached significance (87% of the patients in the patellar tendon group compared with 82% and 75% with normal or nearly normal knee ratings in the hamstring tendon and double-bundle groups, respectively).

The main strength of this study is that it is, to our knowledge, the largest randomized clinical trial reporting the 5-year quality-of-life and clinical outcomes of patients treated with a patellar tendon, hamstring tendon, or double-bundle ACL reconstruction (95% follow-up). One perceived limitation may be the lack of generalizability, with recruitment occurring at a single orthopaedic practice and all surgeries performed by the same surgeon. However, these mid-term results are consistent with those of other published trials^{13,18,19,21}. The 5-year radiographic assessment (in process and to be reported separately) may shed light on the longer-term outcome concerns relating to posttraumatic osteoarthritis progression.

The biggest concern with ACL reconstructive surgery is the prevention of reinjury involving either knee. Patients must be counseled on the risk of reinjury, which was $>20\%$ over 5 years. This risk is likely much higher for younger patients, as demonstrated at the 2-year follow-up²³. The patients with a traumatic reinjury had a younger average age of 22.1 years, compared with those with no reinjury (29.1 years) and compared with the entire group (28.5 years). Therefore, the most important take-home message from this trial is the prevention of reinjury to the operative and contralateral knee. This is even more of an issue

for younger patients who are at a higher risk. Patients with traumatic reinjuries, including partial ACL graft tears and complete traumatic reruptures, were nearly 4 times as likely to have had a hamstring tendon or double-bundle reconstruction. Using a patellar tendon reconstruction in the higher-risk individual does not prevent reinjury. However, we recommend a patellar tendon reconstruction for higher-risk, younger patients.

Conclusions

At 5 years, there was no significant difference in quality-of-life outcomes among the patellar tendon, hamstring tendon, and double-bundle techniques for ACL reconstruction. There were significantly more traumatic reinjuries in the hamstring and double-bundle groups compared with the patellar tendon group. There was a trend toward a higher percentage of normal and nearly normal IKDC grades in the patellar tendon group compared with the hamstring tendon and double-bundle groups. ■

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